The Chain Rule

Past Differentiation...

The differentiation rules we've learned so far:

- **Power Rule:** $\frac{d}{dx}[x^n] = nx^{n-1}$
- **Sum Rule:** $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
- **Difference Rule:** $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$
- **Product Rule:** $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
- **Quotient Rule:** $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

Preliminaries: What We Know So Far

Question: With the tools we have, how do we find $\frac{d}{dx}(x^3+1)^3$?

- **Expand algebraically first:**

$$(x^3+1)^3 = (x^3+1)^2(x^3+1) = (x^6+2x^3+1)(x^3+1) = x^9 + 3x^6 + 3x^3 + 1$$

- **Differentiate term-by-term via Power Rule:**

$$\frac{d}{dx}(x^3+1)^3 = \mathbf{9x^8 + 18x^5 + 9x^2}$$

Question: Does the same idea practically let us find $\frac{d}{dx}(x^3+1)^{100}$?

- **Board Answer:** *Not really! Much too long!* We need a rule specifically for function composition.

Practical Motivation via Units of Change

Let $L(E)$ be the life expectancy (years) of a person with E years of education. Let $E(i)$ be the average number of years of education obtained by people with an average annual income of i dollars.

- **Describe $L(E(i))$ in practical terms:** Life expectancy (in years) of a person whose parents have an annual income of $\$i$.
- **Units of Derivatives:**

$$\frac{dL}{dE} = \frac{\text{years of life expectancy}}{\text{year of education}}, \qquad \frac{dE}{di} = \frac{\text{years of education}}{\$ \text{ of annual income}}$$

- **Chain of Rates:** Multiplying the rates cancels the intermediate unit:

$$\frac{dL}{di} = \frac{dL}{dE} \cdot \frac{dE}{di} = \frac{\text{years of life expectancy}}{\$ \text{ of annual income}}$$

The Chain Rule

If $f(x)$ and $g(x)$ are differentiable functions, then the composite function is differentiable and:

$$\mathbf{\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)}$$

Leibniz Notation: If we set $u = g(x)$ and $y = f(u)$, then $y = f(g(x))$ and:

$$\mathbf{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Example: Differentiating $(x^3+1)^3$ via the Chain Rule

- Outer function: $f(u) = u^3 \implies f'(u) = 3u^2$
- Inner function: $g(x) = x^3 + 1 \implies g'(x) = 3x^2$
- Composition: $f'(g(x)) \cdot g'(x) = 3(x^3+1)^2 \cdot (3x^2) = 9x^2(x^6+2x^3+1) = \mathbf{9x^8 + 18x^5 + 9x^2}$ *(Matches!)*

Proving the Quotient Rule & Multi-Layer Chain Rules

Proving the Quotient Rule Using the Chain Rule

The quotient rule tells us how to differentiate $\frac{f(x)}{g(x)}$. Rather than using difference quotients from scratch, we can prove it naturally by combining the **Product Rule** and the **Chain Rule**.

- **Rewrite as a Product of Powers:**

$$\frac{f(x)}{g(x)} = f(x) \cdot [g(x)]^{-1}$$

- **Step 1: Differentiate the Reciprocal $[g(x)]^{-1}$ via the Chain Rule:**

$$\text{outer: } u^{-1} \implies \frac{d}{du}[u^{-1}] = -u^{-2} = -\frac{1}{u^2}, \qquad \text{inner: } u = g(x) \implies u' = g'(x)$$

$$\frac{d}{dx}\big([g(x)]^{-1}\big) = -[g(x)]^{-2} \cdot g'(x) = -\frac{\mathbf{g'(x)}}{\mathbf{(g(x))^2}}$$

- **Step 2: Differentiate the Product $f(x) \cdot [g(x)]^{-1}$:**

$$\begin{aligned}
\frac{d}{dx}\left[f(x)[g(x)]^{-1}\right] &= f'(x)[g(x)]^{-1} + f(x)\frac{d}{dx}\big([g(x)]^{-1}\big) \\
&= \frac{f'(x)}{g(x)} + f(x)\left(-\frac{g'(x)}{(g(x))^2}\right) \\
&= \frac{f'(x)g(x)}{(g(x))^2} - \frac{f(x)g'(x)}{(g(x))^2} \\
&= \frac{\mathbf{f'(x)g(x) - f(x)g'(x)}}{\mathbf{(g(x))^2}} \qquad \checkmark
\end{aligned}$$

The Triple Chain Rule (and Higher Compositions)

What if a function consists of three or more nested layers, such as $f(g(h(x)))$?

The Triple Chain Rule:

$$\mathbf{\frac{d}{dx}\big[f(g(h(x)))\big] = f'\big(g(h(x))\big) \cdot g'\big(h(x)\big) \cdot h'(x)}$$

General n-Layer Composition: Differentiate from the outermost layer to the innermost layer, evaluating each derivative at the remaining nested inner expression, and multiplying by the derivative of each successive layer.

Example Worked on Blackboard: Differentiating $\big((x^3+1)^7 - 3\big)$

Identify the structural decomposition across the three layers:

Layer	Component Function	Standard Derivative	Evaluated at Inner Argument
Innermost	$h(x) = x^3 + 1$	$h'(x) = 3x^2$	$h'(x) = 3x^2$
Middle	$g(u) = u^7$	$g'(u) = 7u^6$	$g'(h(x)) = 7(x^3+1)^6$
Outermost	$f(v) = v - 3$	$f'(v) = 1$	$f'(g(h(x))) = 1$

Applying the Triple Chain Rule:

$$\begin{aligned}
\frac{d}{dx}\Big[\big((x^3+1)^7 - 3\big)\Big] &= f'\big(g(h(x))\big) \cdot g'\big(h(x)\big) \cdot h'(x) \\
&= 1 \cdot \big[7(x^3+1)^6\big] \cdot \big(3x^2\big) \\
&= \mathbf{21x^2(x^3+1)^6}
\end{aligned}$$